

Tae Yong Song

Ph.D. Student in Building Science

Department of Architecture and Architectural Engineering, Seoul National University

Website | Google Scholar | GitHub | LinkedIn

RESEARCH PROFILE

I study building intelligence at the intersection of building physics, performance simulation, and artificial intelligence. My work focuses on models that remain physically meaningful, transferable, and useful for building-energy decisions when operational data are limited.

EDUCATION

2026–present	Ph.D. in Architecture and Architectural Engineering , Seoul National University Building science and building performance simulation
2022–2023	M.S. in Engineering , Seoul National University Thesis: “Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model”
2015–2021	B.S. in Architecture and Architectural Engineering , Seoul National University

PUBLICATIONS

International Journal Article

- 2026** Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288.

International Conference Papers

- 2026** Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Tae Yong Song, Jin-Hong Kim, Seung-Ju Lee, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.
- 2024** Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.
- 2023** Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.
- 2023** Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

- 2026** Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms
Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Awarded the Outstanding Presentation Paper Award (General Division, Oral Presentation).**
- 2022** Dynamic Sensitivity Analysis Using NODE-GAM
Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

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| 2026–2030 | Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation |
| 2025–2026 | LLM Technology for Digital-Twin-Based Building Energy Models |
| 2025–2026 | IDF Model Development for Digital-Twin Energy Simulation |
| 2022–2024 | Automated Digital Diagnosis and Design Technologies for Green Remodeling |
| 2023 | Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control |
| 2022–2023 | Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling |

TEACHING EXPERIENCE

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| 2026–present | Teaching Assistant , Seoul National University
Department of Architecture and Architectural Engineering |
| 2023 | Teaching Assistant , Seoul National University
Department of Architecture and Architectural Engineering |

HONOR

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| 2026 | Outstanding Presentation Paper Award
Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division |
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